

# Geometry-Aware JEPA for EEG

A controlled frozen-transfer study on TUAB

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Hack the World(s) – EEG Track   ·   Team **Hello Worlds**

## Contribution

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## Our contribution, in one slide

1. **A strong frozen baseline.** SIGReg-ambient JEPA matches the LeJEPA/Laya recipe's value on our setup (pipeline-helped: organizers' preprocessing + eb\_jepa dataloader/losses) – not a controlled head-to-head with Laya.
2. **A broad ablation that does not beat it.** Ambient/tangent SIGReg, ambient/tangent PEIRA, Graph-JEPA, Fourier-JEPA – none improves balanced accuracy beyond noise ( $\leq 0.013$ ). Ambient SIGReg is hard to beat.
3. **Latent geometry is the methods contribution.** Our latent separates normal/abnormal  $\sim 2\times$  better than random on TUAB; we visualise it with **de Surrel's AIRM Riemannian t-SNE** – the principled metric for SPD/EEG covariances – which gives a small, *consistent* edge over classical t-SNE *where cluster structure exists*. Tangent training yields the *best* covariance-latent silhouette (TUAB & TUSZ), though *not* better balanced accuracy – the geometry objective sharpens the geometry, not the task.
4. **Rigor over spin.** We report the limits: *decodable*  $\neq$  *clustered* (TUEV), *sustained* vs *transient*, the AIRM edge within noise. An honest negative beats a fragile positive.

# Motivation

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## The question we actually asked

- **EEG foundation models** are everywhere – but most report *fine-tuned* numbers. What survives when you **freeze** the encoder?
- **Self-supervised JEPA** avoids collapse with an *anti-collapse regulariser*. The EEG covariance lives on a curved **SPD manifold**, not in flat Euclidean space.
- **Our pre-registered hypothesis**: making anti-collapse *geometry-aware* (act in the SPD **tangent**) and *distribution-free* (**PEIRA**) beats a plain ambient baseline.
- **What we ship**: a controlled study with an *honest* answer – a strong frozen probe, a clean negative result with a geometric mechanism, and label/data-efficiency curves.

*We are not a foundation model, not a world model, not SOTA in 24h.*

# The frozen-transfer protocol



- **Why frozen?** It measures the *representation*, not the fine-tuner. Apples-to-apples across methods, and the regime where many published FMs quietly collapse.
- **Selection rule:** probe hyper-parameters are fixed *a priori* (StandardScaler + balanced LogisticRegression, max\_iter=3000) – **no** hyper-parameter search; the 276-recording eval is touched **once** with a train-only scaler – no selection leakage.

# Data

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## Data: TUH Abnormal EEG (TUAB)

- **Task:** binary *normal* vs *abnormal* EEG, recording level.
- **Signal:** 19 channels @ 200 Hz; we sample 10 s windows (2,000 samples), per-channel z-scored.
- **Standard split:** 2,717 train / 276 eval recordings, **patient-disjoint** (2,076 vs 253 patients, overlap = 0 – verified).
- **Pretraining:** SSL reads only *unlabeled* train recordings; labels enter at the probe alone.

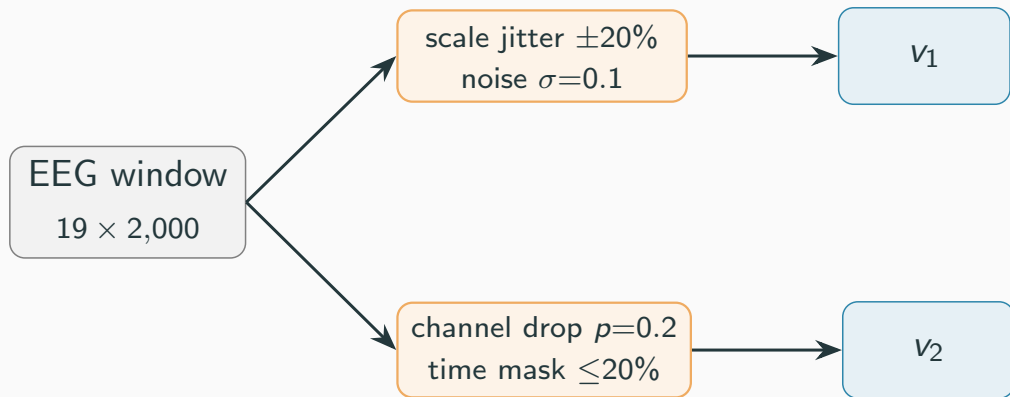
### Why we don't chase accuracy

TUAB labels come from clinical reports and are known to be noisy; we are not aware of a published label-noise / Bayes-ceiling estimate. **Sub-1-point gaps without multi-seed CIs are not meaningful** – so we report *balanced accuracy on the full split*, 3 seeds, with error bars.



## Two views for a JEPA invariance objective

Each SSL sample = one 10 s window, augmented **twice** independently into views  $v_1, v_2$ :

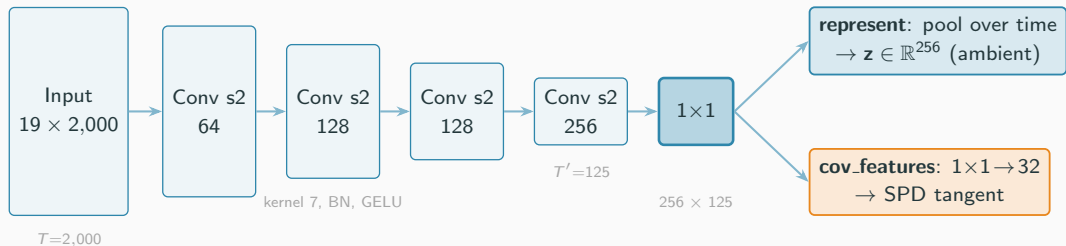


- Augmentations (per channel, in z-scored units): amplitude scale jitter, additive

# Architecture

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## Encoder: a 1D convolutional EEG backbone



4 strided Conv1d blocks (kernel 7, stride 2):  $T : 2,000 \rightarrow 125$ , width  $19 \rightarrow 256$ .

**One backbone, two readouts** – the probe always reads the pooled represent.

# Where can anti-collapse act? Ambient vs SPD-tangent

## Ambient (Euclidean).

Pool the feature map over time:

$$\mathbf{z} = \frac{1}{T'} \sum_t f_{\theta}(x)_{:,t} \in \mathbb{R}^{256}.$$

Regularise directly in  $\mathbb{R}^{256}$ . This is the

**Laya-like** baseline.

## Tangent (SPD log-Euclidean).

Per-window temporal covariance of a 32-dim projection:

$$C = \frac{1}{T'-1} \tilde{F} \tilde{F}^{\top} \in \text{SPD}(32),$$

$$\mathbf{t} = \text{vec}_{\sqrt{2}}(\log_m C) \in \mathbb{R}^{32(32+1)/2}.$$

Regularise in the tangent at the identity.

## The clean knob

The probe reads the *same* pooled representation in both cases. Only the *space where anti-collapse is enforced* changes – this isolates the effect of **geometry**, nothing else.

## The ladder – a $2 \times 2$ (+ reference) factorial

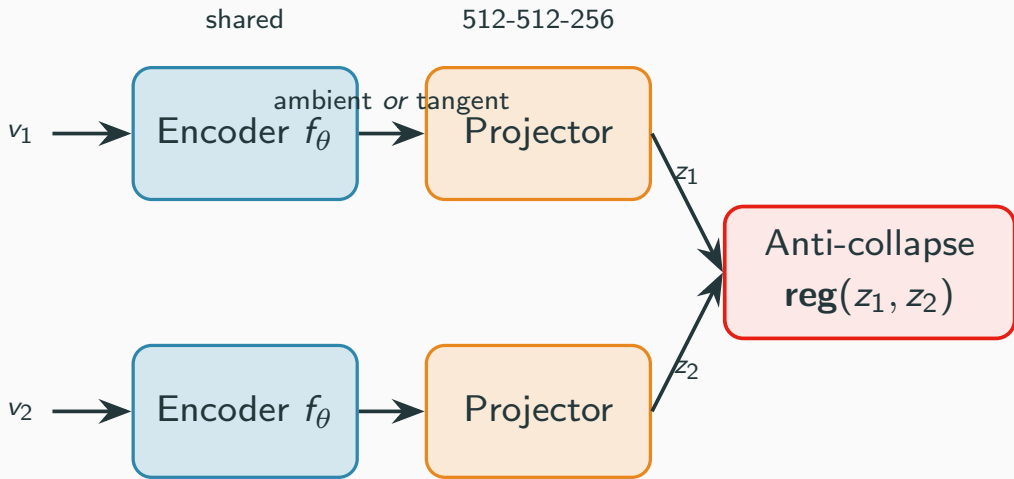
| regulariser \ space                | ambient (Euclidean)          | tangent (SPD)           |
|------------------------------------|------------------------------|-------------------------|
| <b>VICReg</b> (variance/cov)       | C0 reference                 | —                       |
| <b>SIGReg</b> (Gaussianity slices) | C1 <i>Laya-like baseline</i> | C2 geometry-aware       |
| <b>PEIRA</b> (distribution-free)   | C3                           | C4 <i>ex-hypothesis</i> |

- **SIGReg** (LeJEP/BCS): tests random 1-D slices for Gaussianity – assumes an *isotropic* target.
- **PEIRA**: distribution-free anti-collapse – no target distribution to mis-specify.
- **Hypothesis**: the geometry-aware / distribution-free cells (C2, C4) beat the ambient baseline C1.

# Training

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## Two-view JEPA pretraining



- AdamW, lr  $10^{-3}$ , weight decay  $10^{-6}$

- Collapse monitored *every epoch*: effective 9 rank, per-dim std. off-diagonal covariance

# Collapse is the #1 risk – so we watched it

A degenerate encoder maps every window to (almost) the same vector. We log three diagnostics on each batch:

- **Effective rank** – entropy of the singular-value spectrum;  $\rightarrow 1$  means collapse.
- **Per-dim std** –  $\rightarrow 0$  means collapse.
- **Off-diagonal covariance** – redundancy across dimensions.

## Rule

Kill flat runs early. The fix for collapse is *more* regularisation, not less. Every reported cell was checked non-degenerate (high eff. rank; PEIRA's  $tr_P$  rising).

## Jury asset

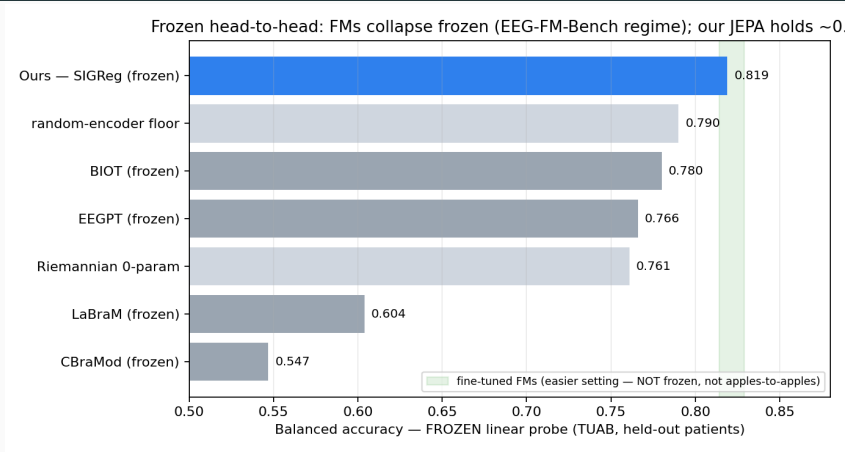
“We didn’t just avoid collapse – we *visualised* it.”



## Results

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# Frozen & in-domain: we hold where frozen FMs fall



Frozen linear probe, TUAB held-out patients. **Ours 0.819** (3-seed) sits above the random floor and every frozen FM baseline; fine-tuned FMs (green band) are an *easier*, non-comparable setting.

# Apples-to-apples: it is not an in-domain artifact

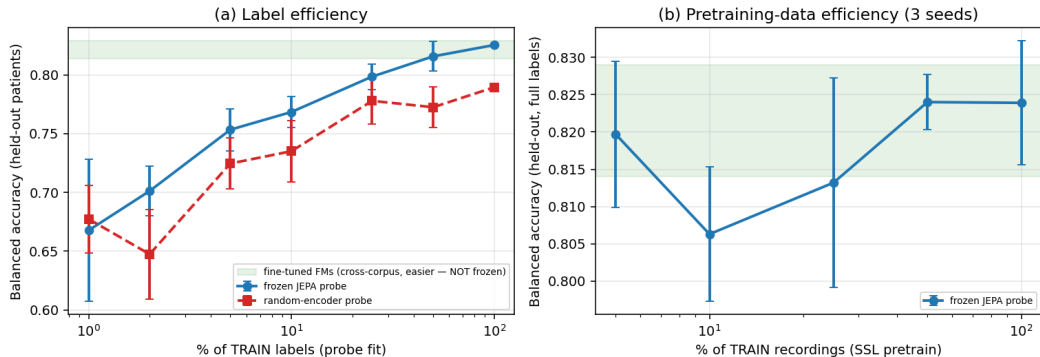
- “*You pretrained on TUAB, the FMs didn’t*” – the obvious attack on the in-domain number. So we ran the fair version.
- Pretrain SIGReg on a **general TUH corpus** (**TUSZ** seizure; TUAB-eval patients *excluded*  $\Rightarrow$  patient-disjoint), freeze, linear-probe TUAB – the FMs’ own regime (general pretrain  $\rightarrow$  frozen TUAB).
- **Result: 0.814 BA / 0.889 AUROC** – essentially equal to in-domain 0.819. **The in-domain advantage is not the driver.**

## Honest caveat

TUSZ is general *same-site* TUH (not multi-site like the FMs); single seed. We say “match while frozen”, never “beat”.

# The value of self-supervision (label & data efficiency)

Value of self-supervision — frozen in-domain EEG-JEPA on TUAB



(a) The money plot: *JEPA @ 25% labels*  $\approx$  *random-encoder @ 100%* —  $\sim 4\times$  label efficiency,  $\sim +0.04$  BalAcc. (b) Probe is stable down to small pretraining-data fractions.

## The honest denominator: a random encoder already scores 0.79

- In our pipeline, a **random (untrained) conv encoder** + linear probe reaches  $\sim 0.79$  BalAcc.
- EEG abnormality is *power-driven*; random conv filters preserve band power. The strong frozen numbers are largely *architecture + task*.
- This floor is  $\approx$  LaBraM-linear (0.795) / EEGPT-frozen (0.798) *as self-reported by those papers* (different protocols); under the one apples-to-apples bench we cite (EEG-FM-Bench) the floor sits *above* all frozen FMs.

### Why disclose it?

It pre-empts “is the SSL doing anything?”, reframes our contribution as a **quantified** SSL increment ( $\sim +0.04$  over random,  $\sim 4\times$  label efficiency) – and is itself a sharp benchmark insight: *TUAB frozen-probe is easy for simple power features – in our consistent linear-probe setup, several published “foundation-model” frozen numbers barely clear this random baseline.*

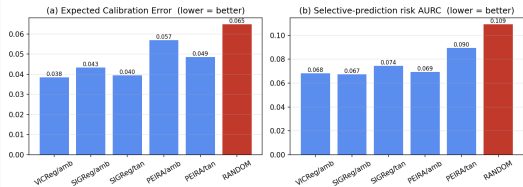
## Where 0.819 sits – TUAB standard 2717/276 split

| Setting                            | Protocol                                   | BA                | AUROC        |
|------------------------------------|--------------------------------------------|-------------------|--------------|
| <b>Ours</b>                        | <b>in-domain SSL, frozen, linear probe</b> | <b>0.819</b>      | $\sim 0.89$  |
| LaBraM-Base / -Huge                | fine-tuned                                 | 0.814 / 0.826     | 0.902 / 0.91 |
| CBraMod                            | fine-tuned                                 | 0.829             | 0.923        |
| REVE-Base                          | fine-tuned                                 | 0.832             | 0.925        |
| LaBraM-Base                        | linear probe                               | 0.795             | 0.884        |
| REVE                               | linear probe                               | 0.810–0.821       | —            |
| EEGPT                              | frozen / linear-probe-style                | 0.798             | 0.872        |
| EEG2Rep                            | linear probe                               | 0.766 (acc)       | 0.832        |
| BIOT                               | linear probe                               | 0.751 (acc)       | 0.829        |
| classical Riemannian (Gemein 2020) | cov + tangent + classifier                 | $\sim 0.86$ (acc) | —            |

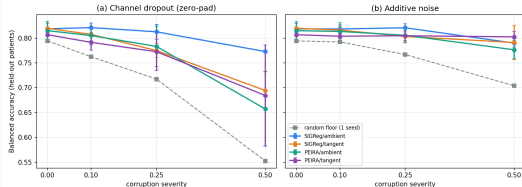
**Top of the frozen linear-probe results;** lower edge of the *fine-tuned* BA band. AUROC  $\sim 0.89$  above BIOT/EEGPT, below CBraMod/REVE. **Competitive at the operating point – not SOTA.**

# Beyond accuracy: calibration & robustness (3 axes, not 1)

Frozen-probe reliability on TUAB — SSL beats the random floor; geometry/PEIRA does not beat ambient (1 seed)



Frozen-probe robustness on TUAB (3-seed mean  $\pm$  std) — ambient SIGReg is the most dropout-robust



- **Calibration:** every SSL encoder beats the random floor (ECE, selective prediction) – geometry/PEIRA does *not* beat ambient.
- **Robustness (3-seed):** ambient SIGReg is the *most* channel-dropout-robust; the SPD-tangent objective *degrades* dropout-robustness; no geometry win under noise (the single-seed hint did not replicate).
- **The null holds on all three axes** – accuracy, calibration, robustness. Geometry buys none of them here.

**Insight**

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## The result we pre-registered against: a clean null

3-seed mean balanced accuracy,  $\{1, 1000, 10000\}$ :

| reg \ space | ambient               | tangent            |
|-------------|-----------------------|--------------------|
| VICReg      | 0.814                 | —                  |
| SIGReg      | <b>0.819</b> (locked) | 0.820 <sup>†</sup> |
| PEIRA       | 0.815                 | 0.807              |

- Every cell  $\approx 0.82 \pm$  seed noise; per-cell ranges overlap heavily. **No effect of regulariser, no effect of space, no interaction.**
- Our hypothesis (*geometry / PEIRA win*) is **falsified**. We *locked* ambient SIGReg and kept the  $2 \times 2$  as the published ablation.

<sup>†</sup> caveat: SIGReg’s slice schedule is seeded per-step, not per-run, so its “tightest variance” is partly an artifact – we do not lean on it (see appendix).

## Why the naive tangent is mis-specified (the mechanism)

On the SPD manifold, the natural Gaussian is a **wrapped Gaussian** – *non-isotropic* by construction (de Surrel et al. 2025, Def. 4.1:  $t \sim \mathcal{N}(\mu, \Sigma)$  with full  $\Sigma$ , not  $\sigma^2 I$ ).

SIGReg forces *isotropy* ( $\mathcal{N}(0, I)$ ) where the geometry induces a non-isotropic covariance – it fights the structure that carries the signal.

|         | ours      | principled             |
|---------|-----------|------------------------|
| target  | isotropic | wrapped, full $\Sigma$ |
| base pt | identity  | Fréchet mean           |

### Honest framing

Future-work *motivation*, not a validated mechanism: at  $n=3$  the tangent effect is null, so our data *cannot* confirm the wrapped-Gaussian prediction.

And the principled object is *already* our Riemannian baseline ( $0.761 \ll 0.82$ ): the linear probe re-derives second-order tangent structure for free, so best case for tangent SSL is a *tie*.

## We measured the geometry – not just hid it

- The probe reads the first-order represent; the SPD-tangent lives *only* in the SSL loss. So we exposed the **learned tangent** to the probe directly.
- Learned-tangent probe: **0.79** BA (SIGReg-tangent 0.790 > its random-tangent floor 0.776) – a *real but weak* second-order signal, **below** the first-order rep (0.82).
- TUAB rewards first-order power; the second-order tangent caps  $\sim 0.78$ – $0.80$  (cf. MENDR 0.78– $0.80$ , Riemann 0.761). **The tangent is not broken – it is capped by the task.**

# What we claim – and what we don't

## DO claim

- Strongest among published *frozen*/linear-probe TUAB baselines; inside the fine-tuned BA band.
- A clean **3-seed null** on geometry/PEIRA, with a geometric mechanism + precedent.
- In-domain SSL is the (disclosed) reason it is strong:  $\sim +0.04$  over random,  $\sim 4\times$  label efficiency.

## DON'T claim

- ~~SOTA on TUAB~~ – AUROC below best; preprocessing differs.
- ~~We beat CBraMod/LaBraM~~ – they are fine-tuned; say “*match while frozen*”.
- ~~Tangent-SPD is useless for EEG~~ – only “no benefit in our TUAB frozen pilot”.
- ~~TUAB is saturated at X~~ – ceiling unknown.

## The boxed claim

Domain-matched JEPA pretraining yields a *frozen* encoder whose linear probe (0.819 BA /  $\sim 0.89$  AUROC, 3 seeds) is competitive with the TUAB *fine-tuning* literature and stronger than typical *frozen-probe* baselines. In a controlled  $2 \times 2$  ablation, neither PEIRA nor a geometry-aware tangent improved over ambient SIGReg.

## Limitations & Future Work

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## Limitations & Future Work

- **Single dataset.** Specialization  $\neq$  generality. The honest generality test is frozen cross-dataset transfer (TUEV/Sleep-EDF) – started, but left as future work rather than over-claimed.
- **Underpowered null.**  $n=3$  seeds; the inter-cell gaps ( $\leq 0.013$ ) are below a plausible eval bootstrap CI ( $\pm 0.02$ – $0.03$ , not yet computed). The honest statement is “*benchmark-resolution-limited*”, not “geometry is dead”.
- **The principled fix.** A wrapped-Gaussian SIGReg on the *channel* covariance, AIRM tangent at the running Fréchet mean –  $\sim 40$  LOC + one re-pretrain, with real eigh-backward numerical risk.
- **Where geometry should pay off.** Robustness, calibration, and transfer under montage/threshold shift – *not* a balanced-accuracy race.

## Credit & honest scope

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# Credit & honest scope: what 24h bought (and what it didn't)

**Built on eb\_jepa – and it carries the baseline.**

- The *EEG dataloader/preprocessing* and the *SSL loss primitives* (VICReg, SIGReg/BCS) are eb\_jepa (the handed framework). The **encoder architecture and the SSL training loop are ours**; our work is an *additive* comparative study on top – geometry/PEIRA cells, reliability axes, latent viz.
- **The signal preprocessing is the organizers'** (19 ch, 200 Hz, bandpass+notch); eb\_jepa's dataloader + loss primitives make TUAB tractable under a *frozen* probe. We claim the *study*, not the raw accuracy.

**We do not claim SOTA.**

- 24h = a frozen linear-probe study on *one* task (TUAB), TUAB-tuned pipeline. Cross-model numbers are **indicative, not controlled re-runs** of others' pipelines.
- Laya (recent LeJEPa SOTA): **29,109 h / 20,940 subjects / 17 montages / 14 tasks** under the *same* frozen contract – a different league of *generality*. Our single-task value  $\neq$  their generalist value.
- **What 24h bought:** a clean multi-axis *negative, healthy-training* evidence it is not a broken run, and a *manifold-aware* latent viz (AIRM Riemannian t-SNE) – a methods contribution, not a leaderboard claim.



## Conclusion

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## Conclusion

- A **controlled frozen-transfer study**, not a leaderboard entry.
- A **strong frozen probe** (0.819 BA,  $\sim 0.89$  AUROC) that holds where published FMs collapse when frozen – with the random-encoder floor honestly disclosed.
- A **clean, pre-registered negative result**: *where* anti-collapse acts (ambient vs SPD-tangent) and *which* regulariser do not change frozen-probe accuracy on TUAB – with a geometric reason why.
- **The null is multi-axis**: geometry/PEIRA help neither accuracy, nor calibration, nor robustness (3 seeds) – and it holds even *general*-pretrained (TUSZ  $\rightarrow$  frozen TUAB, 0.814).
- **Honesty as the headline**: an honest negative beats a fragile positive.

Built on a trimmed eb\_jepa vendor. Team **Hello Worlds**.

## References

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## References i

- Panchavati, S. et al. (2026). *Laya: a LeJEPa approach to EEG via latent prediction over reconstruction*. Preprint. (*our ambient baseline*)
- de Surrel, T., Lotte, F., Chevallier, S. & Yger, F. (2025). *Wrapped Gaussian on the manifold of Symmetric Positive Definite Matrices*. Preprint.
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## References ii

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- Gemein, L. et al. (2020). *Machine-learning-based diagnostics of EEG pathology (Riemannian baseline)*. NeuroImage.

- Bardes, A., Ponce, J. & LeCun, Y. (2022). *VICReg*. ICLR.

# Appendix

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## Appendix: encoder block + readouts

Each backbone block: Conv1d(kernel 7, stride 2)  $\rightarrow$  BatchNorm  $\rightarrow$  GELU. Widths  $19 \rightarrow 64 \rightarrow 128 \rightarrow 128 \rightarrow 256$ ; a  $1 \times 1$  head fixes the channel dim. Time  $T : 2,000 \rightarrow 125$ .

- $\text{represent}(x) = \text{time-pooled feature map} \in \mathbb{R}^{256}$  – the **only** thing the probe ever reads.
- $\text{cov\_features}(x) = \text{a } 1 \times 1 \text{ projection to 32 channels, fed to the tangent map. } 32 < 125 \text{ keeps the covariance full-rank.}$
- Projector for SSL: 512–512–256 MLP (Linear + BN + GELU), discarded after pretraining.



## Appendix: the SPD-tangent construction

From the feature map  $F \in \mathbb{R}^{32 \times T'}$  (centred over time):

$$C = \frac{1}{T'-1} FF^\top + \varepsilon I \in \text{SPD}(32) \quad (\text{temporal covariance})$$

$$\log_m C = U \text{diag}(\log \lambda_i) U^\top \quad (\text{via eigendecomposition})$$

$$\mathbf{t} = \text{vec}_{\sqrt{2}}(\log_m C) \in \mathbb{R}^{32(32+1)/2} \quad (\text{off-diagonals scaled by } \sqrt{2})$$

- The  $\sqrt{2}$  weighting makes the Euclidean norm of  $\mathbf{t}$  equal the Frobenius norm of  $\log_m C$  – the correct isometry for both LE and AIRM *at the identity*.
- At the identity base point, AIRM-log  $\equiv$  Log-Euclidean: the metric is *not* an independent axis. Mis-specification = isotropic target + identity base point (two axes, not three).
- `eigh`-backward numerics:  $\varepsilon$ -clamped eigenvalues, small 32, float-safe; verified finite on GPU in the smoke test.

## Appendix: collapse diagnostics

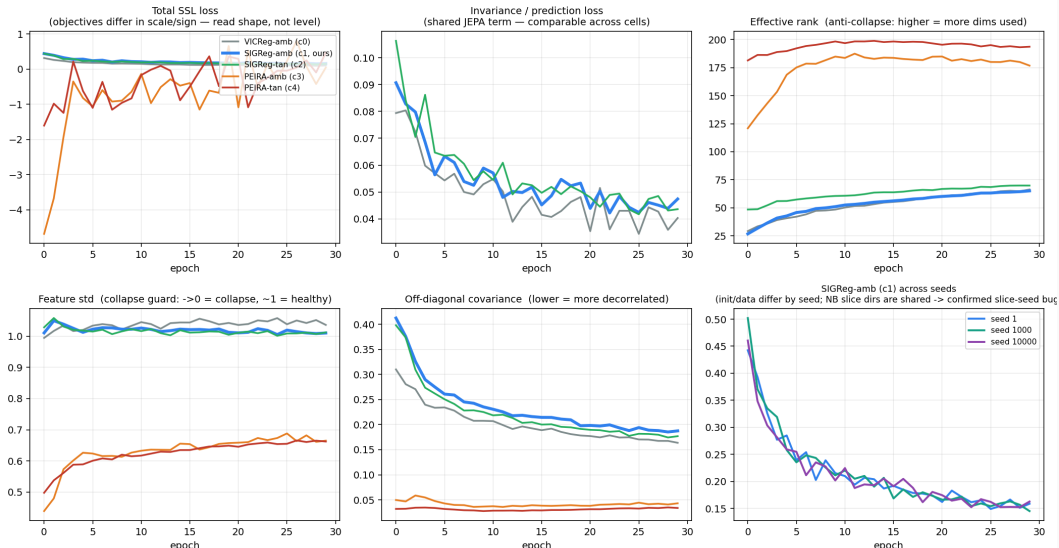
On a batch of embeddings  $Z \in \mathbb{R}^{N \times D}$  (mean-centred), with singular values  $s_i$  and  $p_i = s_i / \sum_j s_j$ :

$$\text{eff. rank} = \exp\left(-\sum_i p_i \log p_i\right), \quad \text{feat. std} = \overline{\text{std}(Z_{:,d})}, \quad \text{off-diag} = \frac{\|\text{Cov}(Z) - \text{diag}\|_F}{\sqrt{D(D-1)}}.$$

- Collapse  $\Rightarrow$  eff. rank  $\rightarrow 1$ , feat. std  $\rightarrow 0$ . Healthy SIGReg-ambient run: eff. rank 27  $\rightarrow$  65 over training.
- For PEIRA, the loss value is a *surrogate-gradient* number (can be negative, non-monotone). Only  $\text{tr}_P$  and eff. rank are valid progress signals.
- Per-cell non-degeneracy was audited before averaging – tangent-PEIRA (0.807) was confirmed healthy, not a partial collapse.

# Appendix: SSL training curves (all cells healthy)

EEG-JEPA SSL training curves — 5 regularizer cells, 30 epochs, TUAB pretrain (seed 1)



## Appendix: evaluation protocol

- **Feature extraction:** 16 evenly-spaced 10 s windows per recording → pooled represent → averaged to a recording-level vector.
- **Probe:** StandardScaler + logistic regression on the frozen features. **Frozen encoder, no fine-tuning.**
- **Split:** train recordings fit the probe; the 276-recording eval (held-out patients) is scored **once**. Probe hyper-parameters are fixed *a priori* (no search); a patient-disjoint dev-split for any future selection is specified but not yet wired in.
- **Metrics:** balanced accuracy (primary) + AUROC, on the full 2,717/276 split.
- **Floor:** an untrained random encoder, same probe, gives the  $\sim 0.79$  honest denominator.

## Appendix: label-efficiency numbers

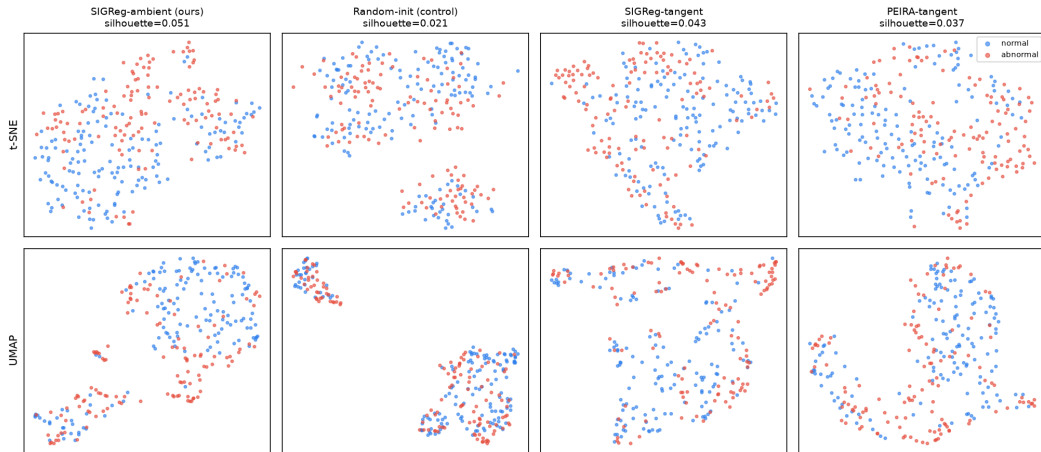
Balanced accuracy vs. % of TRAIN labels used to fit the probe (frozen encoder fixed):

| % labels           | 1     | 2     | 5     | 10    | 25           | 50    | 100          |
|--------------------|-------|-------|-------|-------|--------------|-------|--------------|
| <b>frozen JEPA</b> | 0.668 | 0.701 | 0.753 | 0.768 | <b>0.799</b> | 0.816 | 0.825        |
| random encoder     | 0.677 | 0.647 | 0.725 | 0.735 | 0.778        | 0.773 | <b>0.790</b> |

- **JEPA @ 25% labels (0.799)  $\approx$  random @ 100% (0.790)** – the  $\sim 4\times$  label-efficiency claim.
- The SSL increment ( $\sim +0.04$  at full labels) is small but consistent, and corroborated by a  $\sim 2\times$  higher latent-space silhouette (appendix figure).

# Appendix: frozen latent space (ours vs random vs geometry)

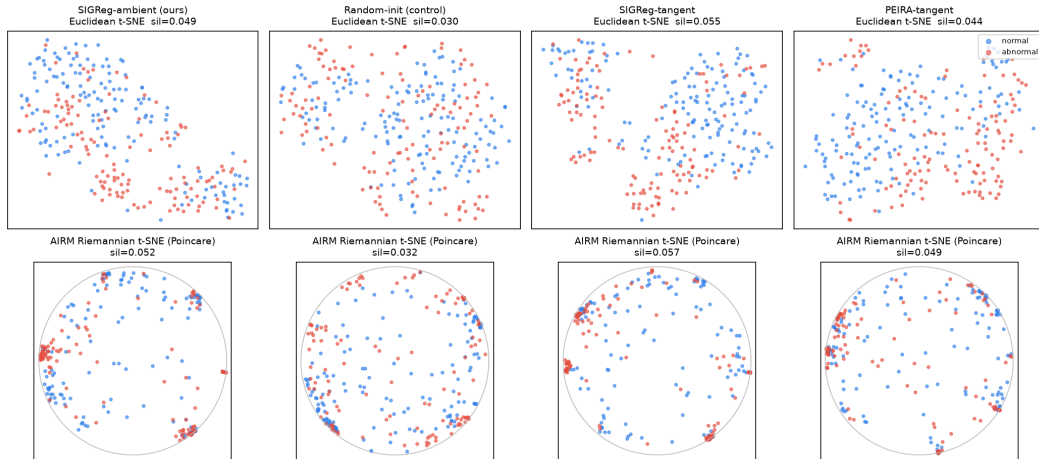
Frozen latent space on TUAB eval — normal vs abnormal (recording-level)  
same frozen ambient represent() readout for all; silhouette is on full-dim features (the 2-D map is illustrative)



t-SNE and UMAP of frozen represent features on the eval set – same frozen ambient readout for all

# Appendix: manifold-aware latent geometry (AIRM vs Euclidean)

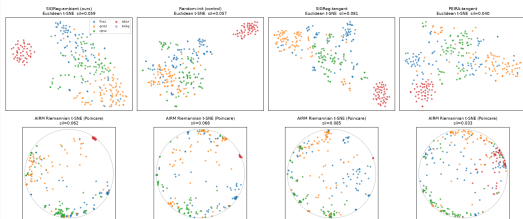
Frozen SPD latents on TUAB eval — Euclidean (tangent) t-SNE vs AIRM Riemannian t-SNE  
silhouette is full-dim (Euclidean tangent vs AIRM geodesic); 2-D maps are illustrative



# Appendix: manifold viz across TUH tasks – sustained vs transient

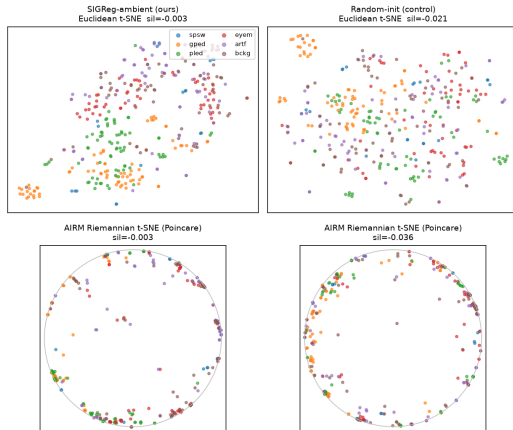
## TUSZ – 5 classes; SIGReg-tangent separates best

Frozen SPD latents on TUSZ eval – 5 seizure types (cross-task, same-site)  
Euclidean (tangent) t-SNE vs AIRM Riemannian t-SNE; silhouette is full-dim



## TUEV – 6 event types (decodable, not clustered)

Frozen SPD latents on TUEV eval – 6 event classes (cross-task, same-site)  
Euclidean (tangent) t-SNE vs AIRM Riemannian t-SNE; silhouette is full-dim



DECODABLE (frozen probe BA -0.40, >> chance 0.17; ours > random) but NOT CLUSTERED (silhouette -0)  
transient sub-second events diluted by the 10 s window -- silhouette measures clustering, not decodability

task (#cls)

sil

silup

- Clusters **sustained** phenomena (abnormality, seizures), not



## Appendix: caveats we disclose to the jury

- **Report 0.819, not 0.833.** The single-seed 0.833 (SIGReg-ambient, seed 1) was the *top* of that cell's range; the 3-seed mean is 0.819.
- **SIGReg slice-seed bug (found, now fixed).** The reported 3-seed SIGReg numbers were produced by code that seeded BCS slice directions from a per-step counter (run-seed-independent), so the 3 seeds shared a byte-identical slice schedule – the error bars *understate* true variance. The code is now fixed (`losses.py` draws slices from the run-seeded RNG); the reported numbers predate the fix and were deliberately not re-run.
- **PEIRA tangent (0.807) is not evidence against wrapped-Gaussian.** PEIRA is distribution-free  $\Rightarrow$  immune to the isotropic-target mis-specification  $\Rightarrow$  it never tests that hypothesis. 0.807 vs 0.815 is within noise.
- **Bootstrap CI (illustrative, not yet run).** A bootstrap over the 276 eval recordings would plausibly give a  $\pm 0.02$ – $0.03$  CI – wider than the max inter-cell gap (0.013) – supporting “benchmark-resolution-limited”. The resampling is not yet committed.
- **Preprocessing differs** from the published FM rows – the head-to-head is indicative, not a controlled re-run of their pipelines.